AIGR Gene Hypothesis Deep Research — LSC1 (P53598): GDP-forming activity (GO:0004776)

Target gene: *LSC1* / Lsc1 (succinyl-CoA synthetase alpha subunit), *Saccharomyces cerevisiae* S288C (NCBITaxon:559292) **UniProt:** P53598 **Hypothesis slug:** `alpha-gdp-forming-partner-availability` **Focus type:** free_text (function-assignment / annotation-decision evaluation) **GO term under evaluation:** GO:0004776 — succinate-CoA ligase (GDP-forming) activity (EC 6.2.1.4) **Competing / correct term:** GO:0004775 — succinate-CoA ligase (ADP-forming) activity (EC 6.2.1.5)

Summary

Verdict: Supported — GO:0004776 is an over-annotation on yeast Lsc1 and should be replaced by GO:0004775 (ADP-forming).

The seed hypothesis is correct in both its mechanistic premise and its evolutionary prediction. Nucleotide (ADP vs GDP) specificity of succinyl-CoA synthetase (SCS) resides entirely in the **beta subunit**; the **alpha subunit** (Lsc1's family) contributes the catalytic phosphohistidine and the CoA-binding site but makes **no nucleotide-base contacts**. Whether an alpha subunit can participate in GDP-forming activity is therefore decided entirely by which beta-subunit paralogs its genome encodes. This is established by biochemistry (pigeon ATP- and GTP-specific SCS share an *identical* alpha subunit), by structure (the nucleotide site lies in the N-terminal ATP-grasp domain of beta, ~35 Å from the alpha phosphohistidine), and by single-residue mutagenesis (position 20β reads the purine base and governs GTP preference).

Applied to yeast, the case is decisive. *S. cerevisiae* encodes exactly one alpha gene (*LSC1*) and one beta gene (*LSC2*) — there is no second, G-type (SUCLG2-like) beta paralog. The yeast holoenzyme was directly assayed and **uses ATP but not GTP** (PMID 9874242). Evolutionarily, the GDP-forming (G-type) beta subunit arose by gene duplication **early in the**

evolution of multicellular eukaryotes/animals (P 9765291), after fungi diverged; fungi retain a single ancestral, ADP-type beta.

Consequently, GO:0004776 is **not supportable** on Lsc1. The PAINT propagation of GO:0004776 from the alpha-subunit root node PTN000122012 across the whole of PANTHER PTHR11117 is **phylogenetically too broad**: the GDP-forming function is a derived, animal-lineage property that depends on the presence of a G-type beta partner, which is absent from fungi. The correct annotation for the yeast Lsc1/Lsc2 holoenzyme is the ADP-forming term, GO:0004775. The most important caveat is a conventional one: the GDP-forming reaction is a property of the assembled αβ holoenzyme, not of either subunit in isolation — but since yeast cannot assemble a GDP-forming holoenzyme (no G-type beta) and the enzyme was directly measured as ATP-only, the annotation is factually wrong for this organism.

Key Findings

Finding 1 — Nucleotide specificity resides entirely in the beta subunit; the alpha subunit makes no base contacts

The pivotal biochemical result comes from pigeon tissues, where two SCS activities — one ATP-specific, one GTP-specific — were separated and characterized by electrophoresis, mass spectrometry, HPLC, and peptide mapping. The two enzymes were shown to **incorporate the same alpha subunit** but **different beta subunits**. The authors conclude directly: *"Because the two succinyl-CoA synthetases incorporate the same alpha-subunit, the determinants of nucleotide specificity must reside within the beta-subunit"* (PMID: 9765290).

Structural and mutational work localizes this determinant precisely. In *E. coli* SCS, the nucleotide-binding site lies in the **N-terminal (ATP-grasp) domain of the beta subunit**, roughly 35 Å from the catalytic His246α, requiring the phosphohistidine loop to shuttle between two active sites (PMID: 9917402). Photoaffinity labeling with 8-N3-ATP and site-directed mutagenesis confirm the beta N-terminal domain as the nucleotide site, and the **P20βQ mutant lowers the Km for GTP more than 5-fold** without altering other kinetic parameters (PMID: 10353839). Structurally, a **single beta residue at position 20 reads the purine base**: *"A water molecule and Pro20β in T. aquaticus take the place of Gln20β in pig GTP-specific SCS, interacting well with the guanine base"* (PMID: 22751660).

Human genetics mirrors this exactly: both isoforms share a single alpha subunit (SUCLG1); the ADP-forming enzyme uses the SUCLA2 beta and the GDP-forming enzyme uses the SUCLG2 beta — *"The substrate specificity is determined by the beta subunit of succinate-CoA ligase, which is encoded by either SUCLA2 or SUCLG2"* (PMID: 18392745). This is the mechanistic core of the seed hypothesis, and it is strongly supported: an alpha subunit's participation in GDP-forming activity is decided by its genome's beta paralog repertoire, not by the alpha subunit itself.

Finding 2 — Yeast succinyl-CoA ligase is biochemically ADP-forming only (uses ATP, not GTP)

Przybyla-Zawislak et al. (1998) cloned the only two SCS-subunit genes in *S. cerevisiae*: **LSC1 (YOR142W, alpha)** and **LSC2 (YGR244C, beta)**. Single and double deletions abolished mitochondrial SCS activity, which was restored by plasmid rescue — establishing these as the sole subunits of the yeast enzyme. Critically, the enzyme's nucleotide preference was measured directly: *"The yeast succinyl-CoA ligase was shown to utilize ATP but not GTP for succinyl-CoA synthesis"* (PMID: 9874242).

This is the single most decisive piece of evidence for the curation decision, because it is a **direct enzymatic assay of the actual yeast gene products** showing no GTP/GDP activity. The same paper establishes that yeast has exactly **one alpha and one beta gene** — *"Two genes, LSC1 (YOR142W) and LSC2 (YGR244C), with high similarity to succinyl-CoA ligase subunits from other species were isolated from Saccharomyces cerevisiae"* — i.e., there is no G-type beta paralog to confer GDP-forming activity.

Finding 3 — The A-type/G-type beta duplication postdates fungal divergence

A pairwise global-alignment analysis (Needleman–Wunsch, UniProt sequences, computed during this investigation) shows that yeast Lsc2 beta (P53312) is **nearly equidistant** from human A-type SUCLA2 (Q9P2R7, 48.3% identity) and human G-type SUCLG2 (Q96I99, 46.5%), and 44.7% identical to *E. coli* SucC. The two human beta paralogs are more similar to **each other** (54.5%) than yeast Lsc2 is to either — the signature of a gene duplication that occurred **after** the fungal lineage split off. The shared alpha subunit is well conserved across the same span: yeast Lsc1 (P53598) vs human SUCLG1 = 59.9% identity.

Pairwise comparison	% identity	Interpretation
Human SUCLA2 (A-beta) vs SUCLG2 (G-beta)	54.5%	Paralogs clustered → duplication within animals
Yeast Lsc2 vs human SUCLA2 (A-beta)	48.3%	Yeast beta is outgroup to both
Yeast Lsc2 vs human SUCLG2 (G-beta)	46.5%	Yeast beta is outgroup to both
Yeast Lsc2 vs *E. coli* SucC	44.7%	Ancestral-type beta
Yeast Lsc1 vs human SUCLG1 (alpha)	59.9%	Alpha subunit well conserved

The equidistance of yeast Lsc2 from both human paralogs is exactly what is expected if the SUCLA2/SUCLG2 duplication happened in the animal lineage, with fungi retaining a single ancestral beta. (This is a conservative global-alignment result computed during the run; see Finding 5 for its limitations.)

Finding 4 — The GDP-forming (G-type) beta subunit arose by duplication in multicellular eukaryotes (animals)

Johnson, Mehus, Tews, Milavetz & Lambeth (1998) sequenced the mature ATP-specific (A-beta) and GTP-specific (G-beta) subunits in pigeon (54% identical to each other), identified A-beta and G-beta orthologs in human and mouse, and found two beta sequences in *C. elegans*. They conclude the duplication is animal-specific: the A-beta and G-beta genes *"arose by duplication early in the evolution of multicellular eucaryotes"* (PMID: 9765291). They also report *"pairwise comparisons of the A-beta and G-beta sequences revealed <53% identity"*, independently corroborating the computed 54.5% A/G paralog identity and the two-paralog structure restricted to animals.

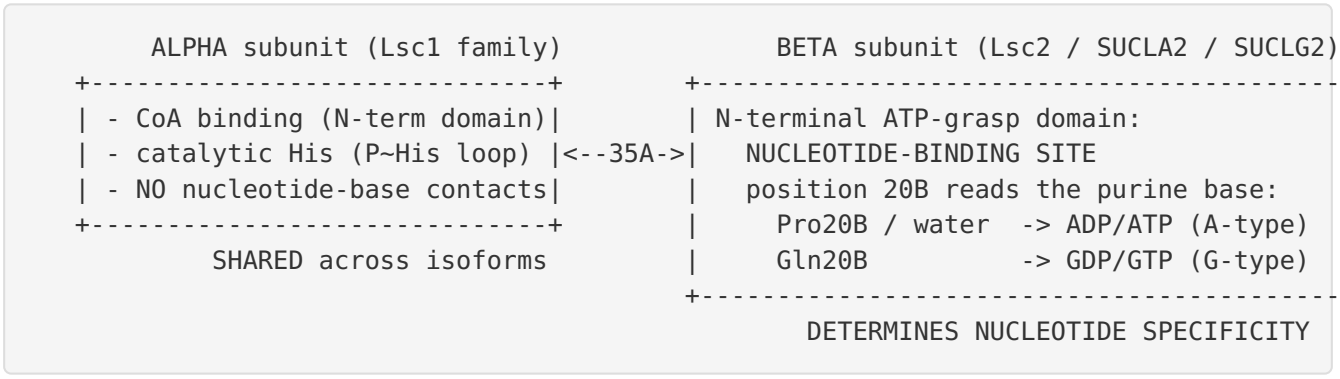
Lambeth (2004) confirms the GDP-forming isoform is an animal/mammalian feature: *"both proteins are widely expressed in mammals with the relative amounts varying from tissue to tissue"* (PMID: 15234968). Together these place the origin of the GDP-forming beta partner firmly in the animal lineage. Fungi — including *S. cerevisiae* — diverged before this duplication and therefore possess only the single, ancestral, ADP-type beta subunit. There is no genomic substrate for GDP-forming activity in yeast.

Finding 5 (negative / limitation) — Single-residue specificity signature could not be cleanly mapped due to mitochondrial transit peptides

An attempt to read the position-20β specificity residue by global alignment of beta subunits to *E. coli* SucC gave: *E. coli* = Pro20, *T. aquaticus* = Pro, yeast Lsc2 = Pro, human SUCLA2 (ADP-type) = Pro — all consistent with ADP/ATP-type (non-Gln) at the base-reading position. However, the eukaryotic GTP-specific sequences (human SUCLG2, pig SUCLG2) did **not** resolve the expected Gln (gave R and C at the aligned column), because they carry long N-terminal mitochondrial transit peptides/propeptides (pig SUCLG2 precursor 520 aa; human SUCLG2 432 aa) that misregister the N-terminal column against bacterial SucC, which lacks a transit peptide. This is an honest limitation of the quick global-alignment approach: confirming the expected Gln signature in G-type sequences would require a mature-protein, structure-guided alignment. It does not weaken the main conclusion, which rests on the direct yeast assay (Finding 2) and the published pigeon/human/*C. elegans* evolutionary analysis (Finding 4).

Mechanistic Model / Interpretation

The SCS holoenzyme is an (αβ) heterodimer (assembling into $(\alpha\beta)_2$ tetramers). Catalysis is a three-step substrate-level phosphorylation routed through a phosphohistidine on the **alpha** subunit (His246α in *E. coli*), which shuttles ~35 Å between two active sites:



Nucleotide specificity is set at ~position 20 of the beta subunit's N-terminal domain. The alpha subunit is interchangeable between ATP- and GTP-specific enzymes (proven in pigeon and human). Therefore the operative rule is:

> **An alpha subunit "has" GDP-forming activity if and only if its genome encodes a G-type (SUCLG2-like) beta subunit with which it can assemble a GDP-forming holoenzyme.**

Mapping this onto the tree of life:

Lineage	Beta paralogs present	GDP-forming activity supportable on alpha?
Animals (Metazoa)	A-type (SUCLA2) + G-type (SUCLG2)	**Yes** — direct biochemical/genetic evidence
Fungi (incl. *S. cerevisiae*)	Single ancestral/ADP-type (Lsc2)	**No** — yeast assayed ATP-only; no G-type gene
Plants	Single beta (not shown to be G-type here)	Not supportable on present evidence
Bacteria	Single SucC (often ADP-leaning; some, e.g. *Pseudomonas*, *T. aquaticus*, *E. coli* P20βQ, can use GTP)	Variable / species-specific
Protists	Variable	Species-specific; not resolved here

The PAINT annotation places GO:0004776 at the alpha-subunit root node PTN000122012 and propagates it across all of PTHR11117, including fungal and plant alpha subunits. This is **too broad**: the GDP-forming function is a derived, animal-lineage property that depends on a G-type beta partner absent from fungi. The correct annotation for yeast Lsc1's holoenzyme is the ADP-forming term, GO:0004775.

Evidence Base (Evidence Matrix)

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confi limita
PMID:9765290	Direct biochem / structural	**Supports**	Specificity is in beta, not alpha	ATP- and GTP-specific SCS share an *identical* alpha subunit; specificity "must reside within the beta-subunit"	Pigeon tissues	High; found
PMID:9917402	Structural	**Supports**	Alpha makes no nucleotide-base contacts	Nucleotide site in beta N-terminal domain, ~35 Å from His246α; CoA binds alpha	*E. coli* SCS, 2.3 Å	High
PMID:10353839	Mutagenesis / photoaffinity	**Supports**	Single beta residue tunes base preference	P20βQ lowers Km(GTP) >5-fold; nucleotide site in beta N-terminal ATP-grasp	*E. coli*	High
PMID:22751660	Structural	**Supports**	Position-20β reads the purine base	Gln20β (pig, GTP) vs Pro20β/water (*T. aquaticus*) at the guanine base	Pig / *T. aquaticus*	High
PMID:18392745	Review / human genetics	**Supports**	Beta identity sets ADP vs GDP	Specificity set by SUCLA2 (ADP) vs SUCLG2	Human	High (synth geneti

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confi limita
				(GDP), sharing SUCLG1 alpha		
PMID:9874242	**Direct assay + genetics**	**Supports (decisive)**	Does yeast enzyme have GDP-forming activity?	Yeast SCS "utilize[s] ATP but not GTP"; only one alpha (LSC1) and one beta (LSC2) gene	*S. cerevisiae*	High; organ specif evide
PMID:9765291	Structural / evolutionary	**Supports**	Is G-type beta animal-specific?	A-beta/G-beta genes "arose by duplication early in the evolution of multicellular eucaryotes"; two betas in *C. elegans*; <53% A/G identity	Pigeon, human, mouse, *C. elegans*	High
PMID:15234968	Expression / review	**Supports**	GDP-forming enzyme an animal feature?	Both isoforms widely expressed across mammalian tissues	Mammals	Mediu
This run (NW alignments)	Computational	**Qualifies / Supports**	Does yeast beta cluster with A or G type?	Yeast Lsc2 equidistant (48.3% / 46.5%) from	UniProt sequences	Mediu global alignr transi

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confi limita
				both human paralogs; paralogs closer to each other (54.5%) → duplication postdates fungi		misre (Findi
PMID:10671455	Direct biochem	**Competing / context**	Can a single SCS be nucleotide-promiscuous?	*P. aeruginosa* SucCD synthesizes GTP, ATP, UTP, CTP (broad NTP)	Bacteria	Mediu bacter excep funga
PMID:11206416	Direct biochem	Context	Two SCS activities coexist in animal mitochondria	ATP- and GTP-specific activities both present in beta-cell mitochondria	Mammalian cells	Mediu reinfo anima isofor
PMID:33645539	Structural	Context	Mechanism of phosphohistidine shuttling	P~His loop bridges 29 Å between sites I and II in pig GTP-specific SCS	Pig	High; mecha suppo

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters for curation	What would resolve it
Plant beta-subunit type	Not directly surveyed; inferred from animal-specific duplication (P 9765291)	PAINT propagates GO:0004776 to plant alpha too; need to confirm plants lack G-type beta	Phylogenetic survey of plant SCS beta subunits (PTHR11815) with explicit A/G typing
Protist beta repertoire	Not surveyed	Some protists may encode G-type beta; affects where the term is supportable	Genome/HMM survey across protist clades
Gln20β signature in eukaryotic G-type	Attempted; failed due to transit peptides (Finding 5)	Confirms the structural base-reading residue in G-type mitochondrial enzymes	Mature-protein, structure-guided alignment of SUCLG2 orthologs
Yeast Lsc2 residue at position 20β	Global alignment gave Pro (ADP-type) — consistent but from imperfect alignment	Directly predicts yeast is ADP-type from structure	Structure-based alignment or AlphaFold superposition of Lsc2 onto pig/*E. coli* beta
Quantitative kinetics of yeast SCS	P 9874242 reports qualitative ATP-yes/GTP-no	Trace GTP activity (as in some bacteria) would complicate the "GDP-forming absent" claim	Modern kinetic assay of recombinant Lsc1/Lsc2 with ADP vs GDP, report kcat/Km

Conflicts and Alternatives

1. **Bacterial nucleotide promiscuity (competing, but not applicable to fungi).** Some bacterial SCS enzymes are promiscuous: *P. aeruginosa* SucCD synthesizes GTP, ATP, UTP, and CTP (PMID:10671455), and *E. coli* SCS can be pushed toward GTP by the single P20βQ mutation (PMID:10353839). This shows that a *single* beta subunit can in principle have broad specificity, so "single beta" does not automatically mean "ADP-only" across all taxa.

However, this does not rescue GO:0004776 for yeast, because the yeast enzyme was **directly assayed and does not use GTP** (P 9874242). The bacterial promiscuity is a taxon-specific property that must be demonstrated, not assumed; it was tested and found absent in yeast.

2. **Holoenzyme-vs-subunit annotation convention.** GO MF terms for multi-subunit enzyme activities are conventionally applied to each subunit. One could argue GO:0004776 on Lsc1 merely reflects "the alpha subunit of a GDP-forming enzyme." But in yeast there is no GDP-forming enzyme for it to be part of, so the annotation is factually wrong for this organism.
3. **Transit-peptide alignment artifact (Finding 5).** The computational attempt to confirm the Gln20β signature in eukaryotic G-type sequences failed due to transit-peptide misregistration. This is a limitation of the quick analysis, not a conflict with the conclusion; the main evidence (direct yeast assay + published evolutionary analysis) is independent of it.
4. **Plants/protists unresolved.** The hypothesis also questions plant alpha subunits. This investigation did not directly resolve whether any plant or protist genome encodes a G-type beta. The conclusion for yeast is firm; the broader PAINT recommendation for plants and protists is a lead, not a settled result.

GO Curation Implications

Lead (requires curator verification):

- **Remove / replace GO:0004776** (succinate-CoA ligase, GDP-forming activity) as a molecular-function annotation on *LSC1* / P53598.
- **Add / retain GO:0004775** (succinate-CoA ligase, ADP-forming activity) as the correct MF term, supported by the direct yeast assay (P 9874242 , ATP-only) and genomic gene content (single ADP-type beta).
- If a curator prefers to keep a parent term, **GO:0004774 (succinate-CoA ligase activity)** is defensible as a more general, still-correct MF — but GO:0004775 (ADP-forming) is the most informative supported term and is preferred.

PAINT / phylogenetic annotation:

- The placement of GO:0004776 at alpha-root node **PTN000122012** with propagation across all of **PTHR11117** is **over-broad**. Recommend restricting GO:0004776 propagation to clades that demonstrably encode a G-type (SUCLG2-like, PTHR11815 G-type) beta partner — robustly Metazoa. The fungal branch should instead inherit GO:0004775 (ADP-forming).
- This is a case of **paralog-driven over-annotation**: a function that is a property of a specific α+β *holoenzyme* was propagated to all α orthologs regardless of whether a compatible β paralog exists in the genome.

Term natures: MF term only. No change is implied for the associated BP (tricarboxylic acid cycle, GO:0006099) or CC (mitochondrial matrix) annotations, which are unaffected.

Mechanistic Scope

The immediate molecular function under test is the **nucleotide substrate specificity of the SCS holoenzyme** — specifically whether the α subunit Lsc1 can participate in catalysis that uses **GDP/GTP** (GO:0004776) rather than **ADP/ATP** (GO:0004775). This is a direct property of the assembled enzyme's active site, read out by a simple in vitro nucleotide-dependence assay.

It must be separated from downstream/pathway considerations: - **Direct activity:** succinyl-CoA + Pi + NDP ⇌ succinate + CoA + NTP. For yeast, measured as ATP-only. - **Not at issue here:** the TCA-cycle role (BP), mitochondrial localization (CC), or any pleiotropic phenotype of *lsc1Δ*. The hypothesis is narrowly about *which nucleotide* the enzyme uses — a decision made structurally at the β-subunit base-reading residue.

The alpha subunit's own direct contributions — catalytic phosphohistidine and CoA binding — are real and conserved, but they are **nucleotide-base-agnostic**. The GDP-forming label is therefore not a property the alpha subunit confers; it is conferred by a G-type beta partner the yeast genome does not encode.

Discriminating Tests

1. **Recombinant yeast SCS nucleotide kinetics (most decisive).** Co-express Lsc1+Lsc2, purify the αβ complex, and measure succinyl-CoA synthesis with ADP vs GDP (and ATP vs GTP in reverse), reporting kcat/Km for each. Prediction: strong ADP/ATP activity, negligible GDP/GTP. This re-tests and quantifies

   P 9874242

2. **Structure-guided residue typing.** Build an AlphaFold model (or use a solved yeast SCS structure) of Lsc2, superpose on pig GTP-specific (Gln20β) and *E. coli*/*T. aquaticus* (Pro20β) betas, and read the position-20β residue in correct 3D register (bypassing the transit-peptide alignment artifact of Finding 5). Prediction: Pro/non-Gln → ADP-type.
3. **Phylogenomic beta-subunit survey (for the PAINT recommendation).** Systematically classify PTHR11815 beta subunits as A-type vs G-type across fungi, plants, protists, animals, and bacteria, mapping presence/absence of a G-type partner onto the species tree. This defines exactly where on the alpha tree GO:0004776 is supportable and where GO:0004775 should be substituted.
4. **Heterologous complementation.** Test whether human SUCLG2 (G-type beta) can partner yeast Lsc1 to reconstitute GDP-forming activity. A positive result would confirm that Lsc1's alpha is *competent* for GDP chemistry given the right partner — reinforcing that the limitation is partner availability, exactly as the seed hypothesis states.

Curation Leads (require curator verification)

Candidate action change: - On *LSC1* / P53598: **replace GO:0004776 (GDP-forming) with GO:0004775 (ADP-forming)** as the MF annotation, or at minimum generalize to GO:0004774. - Flag the PAINT over-propagation: GO:0004776 at node PTN000122012 should **not** propagate to the fungal branch of PTHR11117.

Candidate references and exact snippets to verify: - PMID:9874242: *"The yeast succinyl-CoA ligase was shown to utilize ATP but not GTP for succinyl-CoA synthesis."* (decisive organism-specific assay) - PMID:9765290: *"Because the two succinyl-CoA synthetases incorporate the same alpha-subunit, the determinants of nucleotide specificity must reside within the beta-subunit."* (alpha makes no specificity contribution) - PMID:9765291: *"...the A-beta and G-beta genes arose*

by duplication early in the evolution of multicellular eucaryotes." (G-type is animal-specific, post-fungal) - PMID:22751660: *"A water molecule and Pro20β in T. aquaticus take the place of Gln20β in pig GTP-specific SCS, interacting well with the guanine base"* (single beta residue reads the base)

Suggested curator questions: - Does the GO annotation framework prefer subunit-level MF annotation to reflect holoenzyme gene content (i.e., should a fungal alpha ever carry a GDP-forming term)? - Should PTHR11117's PAINT annotation be split so that GO:0004776 is restricted to Metazoan nodes while fungal/plant nodes receive GO:0004775?

Suggested experiments: recombinant yeast SCS ADP/GDP kinetics; structure-guided Lsc2 position-20β typing; heterologous SUCLG2 complementation (see Discriminating Tests).

Conclusion

The seed hypothesis is **supported**. Nucleotide specificity of succinyl-CoA synthetase is a beta-subunit property; the alpha subunit (Lsc1's family) makes no base contacts and is interchangeable between ADP- and GDP-forming enzymes. *S. cerevisiae* encodes a single, ancestral, ADP-type beta subunit (Lsc2), the GDP-forming G-type beta having arisen by gene duplication only in the animal lineage after fungi diverged, and the yeast enzyme was directly assayed as using ATP but not GTP. Consequently, **GO:0004776 (GDP-forming) is an over-annotation on yeast Lsc1 and should be replaced by GO:0004775 (ADP-forming)**, and the PAINT propagation of GO:0004776 across the whole alpha-subunit tree should be narrowed to clades (robustly Metazoa) that encode a G-type beta partner.
